

QCE Chemistry Units 3&4

Paper 1

Student's Name: _____

Teacher's Name: _____

Time allowed

- Perusal time – 10 minutes
- Working time – 90 minutes

General instructions

- Answer all questions in this question and response booklet.
- QCAA-approved calculator permitted.
- Formula and data booklet provided.
- Planning paper will not be marked.

Section 1 (25 marks)

- 25 multiple choice questions

Section 2 (35 marks)

- 8 short response questions

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2021 QCE Chemistry Units 3&4 Written Examination.

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SECTION 1**Instructions**

- Choose the best answer for Questions 1–25.
- This section has 25 questions and is worth 25 marks.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

	A	B	C	D
Example:	☒	☐	☐	☐

	A	B	C	D
1.	☐	☐	☐	☐
2.	☐	☐	☐	☐
3.	☐	☐	☐	☐
4.	☐	☐	☐	☐
5.	☐	☐	☐	☐
6.	☐	☐	☐	☐
7.	☐	☐	☐	☐
8.	☐	☐	☐	☐
9.	☐	☐	☐	☐
10.	☐	☐	☐	☐
11.	☐	☐	☐	☐
12.	☐	☐	☐	☐
13.	☐	☐	☐	☐
14.	☐	☐	☐	☐
15.	☐	☐	☐	☐
16.	☐	☐	☐	☐
17.	☐	☐	☐	☐
18.	☐	☐	☐	☐
19.	☐	☐	☐	☐
20.	☐	☐	☐	☐
21.	☐	☐	☐	☐
22.	☐	☐	☐	☐
23.	☐	☐	☐	☐
24.	☐	☐	☐	☐
25.	☐	☐	☐	☐

SECTION 2**Instructions**

- Write using black or blue pen.
- If you need more space for a response, use the additional pages at the back of this booklet.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. See page ...
 - If you do not do this, your original response will be marked.
- This section has eight questions and is worth 35 marks.

QUESTION 26 (4 marks)

An aqueous solution containing 1.0 M copper(II) nitrate and 1.0 M magnesium nitrate was electrolysed using inert electrodes. It was found that pure copper metal was deposited on one electrode.

- a) At the other electrode, the pH of the electrolyte near the electrode decreased as the electrolysis proceeded.

Using a balanced half-equation, explain why this happened.

[2 marks]

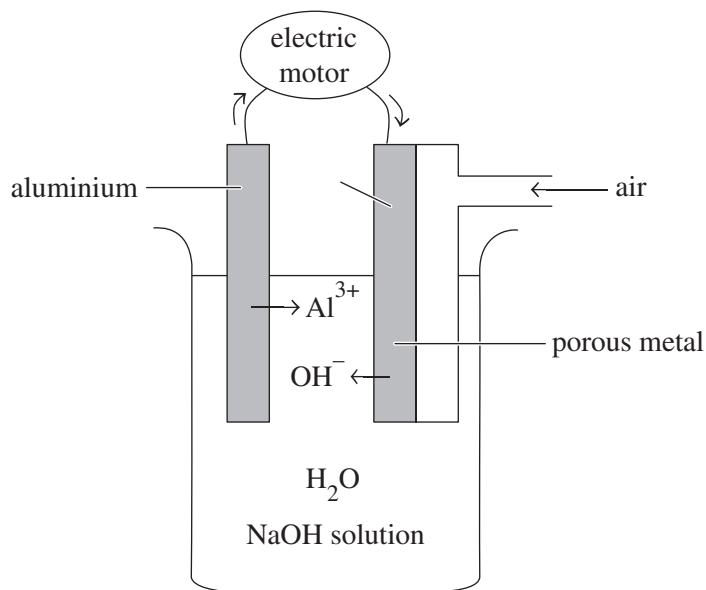
- b) No magnesium metal was deposited during the electrolysis.

Apply your understanding of the standard reduction potentials table to explain why magnesium metal could **not** be deposited in this type of electrolysis experiment.

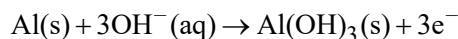
[2 marks]

QUESTION 27 (4 marks)

The aluminum–air battery is a promising power source for electric motor cars. The main features of the battery are shown below.



When the cell operates, the aluminum loses electrons and dissolves, forming aluminum hydroxide. The aluminum plate must be periodically replaced. One reaction in this cell is represented by the following half-equation.



- a) Identify the polarity (positive or negative) of the aluminium electrode. [1 mark]

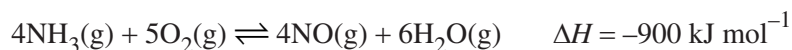
- b) Write a balanced half-equation for the reaction occurring at the surface of the porous metal where oxygen is reduced. [1 mark]

- c) i) Identify one way in which the aluminum–air battery resembles a fuel cell. [1 mark]

- ii) Identify one way in which the aluminum–air battery differs from a fuel cell. [1 mark]

QUESTION 28 (6 marks)

Ammonia gas reacts with oxygen gas according to the following chemical equation.



- a) In one experiment at a particular temperature, 0.300 mol of ammonia gas was mixed with 0.400 mol of oxygen gas in a sealed 2.00 L container and was allowed to reach equilibrium. 0.200 mol of nitrogen oxide gas was present in the equilibrium mixture. Calculate the amount (in mol) of NH_3 , O_2 and H_2O in the equilibrium mixture. Show your working. [3 marks]

Amount of NH_3 at equilibrium = _____ mol (to three significant figures)

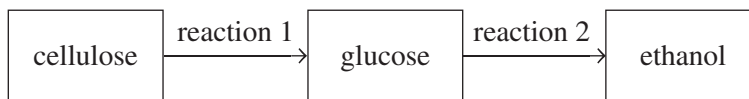
Amount of O_2 at equilibrium = _____ mol (to three significant figures)

Amount of H_2O at equilibrium = _____ mol (to three significant figures)

- b) Apply your understanding of Le Châtelier's principle to predict the effect that an increase in pressure would have on the position of equilibrium for this reaction. Explain your reasoning. [3 marks]

QUESTION 29 (4 marks)

The flow chart below shows a pathway for the production of ethanol.



- a) i) Identify a likely source of cellulose. [1 mark]

- ii) Identify the type of chemical reaction labelled as reaction 1 in the flow chart above. [1 mark]

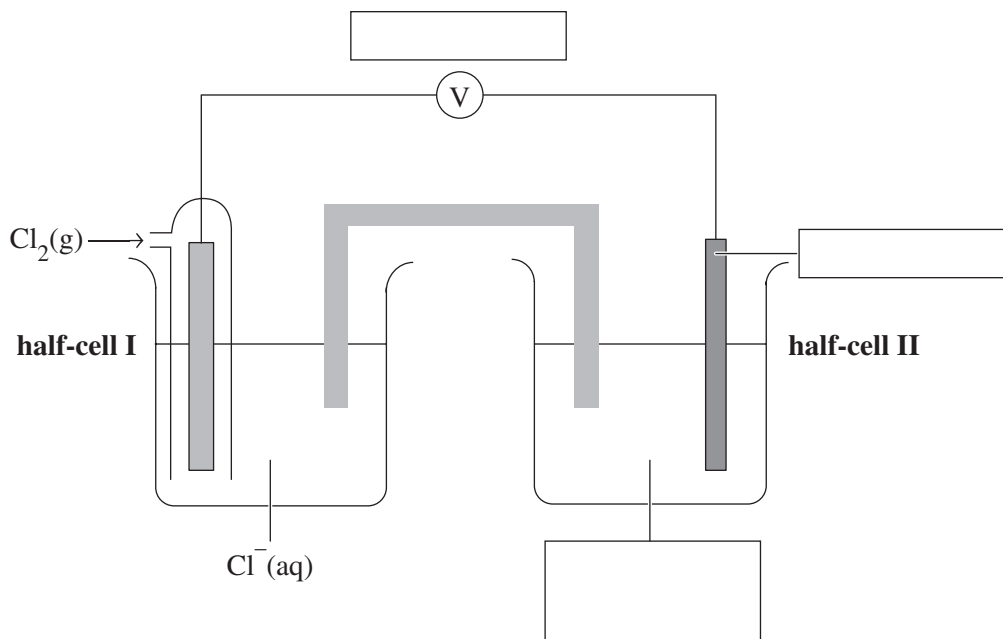
- b) Reaction 2 is known as fermentation and occurs in the presence of a yeast enzyme. Describe two characteristics of enzymes. [2 marks]

QUESTION 30 (4 marks)

Using equations, describe a reaction pathway for the two-step conversion of 1-bromopropane ($\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$) to 1-butanamine. Use semi-structural formulas for organic compounds and include inorganic reagents in your response.

QUESTION 31 (5 marks)

A student wanted to construct a galvanic cell under standard conditions with cell voltage of exactly 0.84 V, using the Cl_2/Cl^- half-cell. A partly labelled diagram of the planned cell is shown below.



- a) Calculate the standard reduction potential (E°) for half-cell II. [1 mark]

Standard reduction potential = _____ V (to two significant figures)

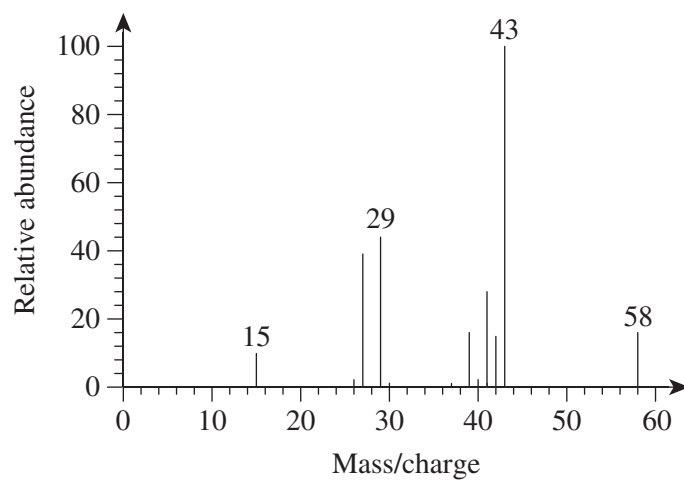
- b) Complete the diagram above by adding the following features. [3 marks]

- an arrow in the box directly above the voltmeter to indicate the direction of movement of electrons
- the composition of the electrolyte in half-cell II
- the composition of the electrode in half-cell II

- c) A salt bridge can be made using a piece of filter paper soaked in a particular liquid. Identify **one** property of the liquid that is required for its use in making a salt bridge. [1 mark]

QUESTION 32 (4 marks)

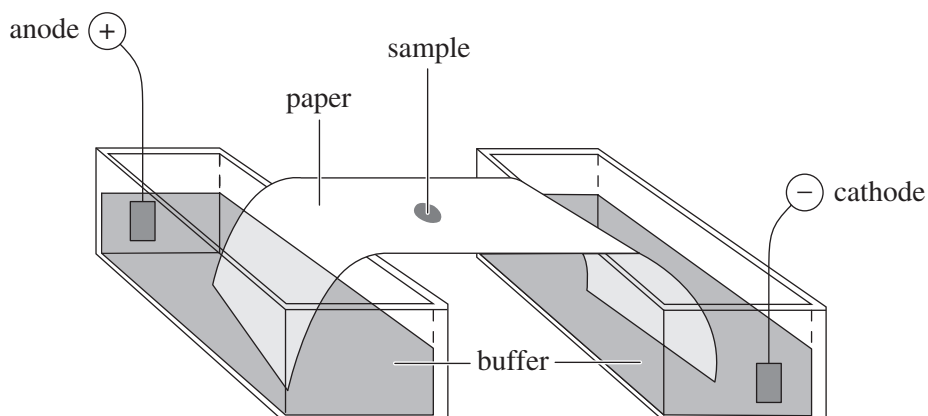
The simplified mass spectrum of a compound with empirical formula C_2H_5 is shown below.



Analyse the spectrum to deduce the semi-structural formula for the molecule. Explain your reasoning.

QUESTION 33 (4 marks)

Electrophoresis can be used to separate a mixture of amino acids containing valine. A sample of the mixture for separation is placed on a paper and an electric field is applied. This set-up is shown in the diagram below.



Valine migrates towards the anode (+) if the buffer solution has a high pH, and towards the cathode (–) if the buffer solution has a low pH.

Explain why this occurs.

END OF PAPER

